

Probabilistic Occupancy Detection and Lighting Control in a Smart Building

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Introduction

This report presents a detailed description of a probabilistic graphical modeling approach for detecting room occupancy and controlling lighting in a smart building environment. The core methodology leverages Hidden Markov Models (HMMs) to capture the temporal dynamics of occupancy states in each room, informed by a diverse set of heterogeneous sensors. The objective is to derive accurate occupancy inferences and translate these into intelligent lighting actions that balance energy efficiency with productivity. The design and implementation are informed by both the characteristics of the sensor data and operational constraints of realistic smart buildings.

Algorithm Description

The occupancy detection problem is framed as a temporal state estimation task, where each room's occupancy is represented as a latent binary variable evolving over time. To capture this, a separate Hidden Markov Model is constructed for each room. The hidden state variable in each HMM corresponds to the room being either *occupied* or *unoccupied*. This latent state evolves according to a Markov process defined by transition probabilities, which model the likelihood of occupancy state changes between consecutive time steps.

Transition probabilities for each room are empirically learned from historical data by counting observed state transitions and normalizing these counts to form stochastic matrices. This data-driven estimation ensures that the model captures realistic occupancy dynamics particular to each room, reflecting patterns such as typical durations of occupancy and frequency of transitions.

A key consideration in configuring HMMs is the choice of initial start-state probabilities. One simple approach is to use the occupancy states at the first timestep from the two training days and average them, providing a crude prior. However, this estimate is extremely noisy due to the small sample size. More robust alternatives include computing the marginal occupancy across all timesteps, estimating the stationary distribution of the room's Markov chain, or incorporating time-of-day priors. In this project, for simplicity, the start-state probability is computed as the ratio of the number of occupancy-change

events (off-diagonal transitions) to the total number of timesteps across both training days. This provides a rough estimate of the frequency of occupancy changes, which, while not a precise representation of the occupancy at $t = 0$, offers a computationally straightforward method to initialize the HMM and ensure the forward algorithm has a starting distribution.

Sensor observations, serving as emissions in the HMM framework, include motion sensors, door counters, camera detections, WiFi presence, and robot detections. These sensors vary in modality and reliability, and their raw signals are preprocessed into binary indicators representing the presence or absence of activity. This binarization process simplifies heterogeneous sensor readings into a unified format conducive to probabilistic modeling.

Custom emission factors are created for each room, reflecting the assumption that only a subset of all sensors is relevant for occupancy detection in that room. Motion sensors are assumed to primarily indicate activity within the room they are located in, though in reality transient movement through the room may introduce noise. Cameras follow a similar assumption, capturing motion primarily in their field of view. WiFi sensor signals are treated as localized, assuming that the presence of a device in a room corresponds to connection with that room’s network, although in practice signals may propagate to nearby rooms. Door sensors are simplified by selecting the closest sensor that a person would reasonably pass to reach a room, based on an educated guess informed by the floor plan. These design choices reduce dependency complexities while retaining the most relevant information for each room.

The emission probabilities, representing the conditional likelihood of sensor readings given the occupancy state, are learned from training data by counting co-occurrences of sensor activations with known occupancy states. The resulting emission factors model the joint distribution $P(\text{sensor readings} \mid \text{state})$ for each room. This approach facilitates sensor fusion by probabilistically integrating evidence across multiple sensors, increasing robustness against individual sensor inaccuracies or failures.

Inference within each room’s HMM employs the forward algorithm. Given N as the number of hidden states (here $N = 2$) and T as the number of timesteps, the forward algorithm has a time complexity of $O(N^2T)$ per room. Since N is constant in this binary occupancy model, the complexity per room is effectively $O(T)$. With $R = 34$ independent rooms processed in parallel or sequence, the overall complexity becomes $O(RT)$, which is tractable for real-time inference even on modest computational resources. The modular nature of the design allows scaling to larger buildings by simply adding more independent HMM instances without affecting per-room runtime.

The decision to turn the lights on or off for each room is determined using an expected cost-based approach. At each time step, the belief distribution of the Hidden Markov Model (HMM) is used to estimate the probability that a given room is occupied, denoted as $P(\text{occupied})$. This probability is extracted from the current belief state of the room.

In addition, the model maintains an estimate of the expected number of people present in the room if it is occupied, denoted as $E[\text{people} \mid \text{occupied}]$. Using these two quantities, the expected cost of keeping the lights off is computed as follows:

$$\text{cost}_{\text{off}} = P(\text{occupied}) \times E[\text{people} \mid \text{occupied}] \times 4$$

The constant factor of 4 represents the cost per person of leaving the lights off when the room is occupied. This cost can be interpreted as the penalty associated with productivity loss due to insufficient lighting.

Once the expected cost is computed, the decision rule is applied:

$$\begin{aligned} \text{If } \text{cost}_{\text{off}} > 1 &\Rightarrow \text{Turn lights ON} \\ \text{Else} &\Rightarrow \text{Turn lights OFF} \end{aligned}$$

Here, the threshold value of 1 represents the break-even point at which the cost of keeping the lights off outweighs the energy savings from turning them off.

Additionally, the system integrates an override mechanism based on robot detections. These detections, which are assumed to be completely accurate indicators of room occupancy, completely ignore probabilistic inference and sets the room’s lighting accordingly.

Justification of Design Choices

A major design decision was to train thirty-four separate HMMs, one for each room, rather than building a single joint HMM for the entire building. The motivation was computational tractability. This approach comes with the assumption that the occupancy of a room does not depend on the occupancy of any other room. While this is not true in the real world—where total building occupancy is often normally distributed and rooms influence one another—it greatly simplifies the problem. Accounting for inter-room influence would require a single HMM where each hidden state encodes the occupancy configuration of all 34 rooms. In the simplest binary case, this results in 2^{34} possible states, which is computationally infeasible given the available resources. Therefore, the chosen approach of using 34 simpler, independent HMMs is a pragmatic trade-off between accuracy and feasibility.

The choice of Hidden Markov Models as the foundational modeling technique is motivated by the intrinsic temporal nature of occupancy patterns. Occupancy states evolve with temporal dependencies, and HMMs provide a mathematically principled framework to model such sequential stochastic processes. By explicitly modeling state transitions, HMMs accommodate the persistence and change of occupancy over time, which static classifiers or snapshot models fail to capture effectively.

Binarization of sensor data standardizes inputs from diverse sensor types, reducing the complexity and noise inherent in raw signals. Custom emission factors further enhance this approach by tailoring the probabilistic model for each room, reflecting domain knowledge from the floor plan and likely occupancy paths. This design prioritises the most

relevant sensor inputs, improving inference accuracy while controlling computational demands.

Assumptions and Realism

Several key assumptions underpin the modeling framework. The Markov assumption posits that the occupancy state at each timestep depends only on the previous state, ignoring longer temporal dependencies. While this is a simplification, it is justified by typical room usage patterns, where occupancy tends to change gradually rather than abruptly.

Sensor reliability is assumed to be sufficient for effective binarization and probabilistic modeling. Although sensors are noisy and may produce false positives or negatives, the fusion approach mitigates individual sensor errors. Robot detections are assumed to be highly accurate and timely, an assumption reasonable given their targeted deployment.

The independence of room occupancy models neglects possible correlations, such as adjacent rooms being occupied simultaneously. This assumption is primarily for computational tractability and modularity, with potential to be relaxed in future work through coupled models or graphical structures encoding spatial dependencies.

Custom emission factors rely on assumptions about the relevance of each sensor to each room. Motion sensors and cameras are assumed to primarily indicate activity within their respective rooms. WiFi signals are assumed localized to a single room, although in reality, signal propagation may occur. Door sensors are simplified by selecting the most likely sensor influencing each room, an approximation guided by floor plan interpretation and expected movement paths. While these assumptions are simplifications of complex real-world dynamics, they are grounded in plausible human behavior and spatial considerations.

Expected people counts per room are treated as static, derived from the training data and assumed stable over time. This ignores potential temporal variations such as different occupancy patterns on more work intensive days for instance.

Alternative Strategies

Alternative modeling approaches could include Dynamic Bayesian Networks or Coupled HMMs to explicitly capture dependencies between rooms and model multi-occupancy states beyond binary presence. These richer models might improve accuracy but at the expense of increased computational complexity and data requirements.

Incorporating continuous sensor data, rather than binarizing, could preserve more information and enable finer-grained inference. Advanced sensor models capturing sensor noise characteristics or employing deep learning methods for feature extraction represent promising directions.

Conclusion

This project demonstrates a rigorous application of probabilistic graphical models to the practical problem of occupancy detection and lighting control in smart buildings. The use of room-specific Hidden Markov Models leverages temporal dynamics and sensor fusion to robustly infer occupancy from noisy heterogeneous data. Custom emission factors incorporate domain knowledge about sensor relevance based on the floor plan, and robot-based overrides ensure safety and responsiveness beyond probabilistic inference. The design incorporates realistic assumptions about sensor reliability, occupant behavior, and building operation, while providing interpretable control decisions grounded in expected occupant impact. Future work could extend the model's expressiveness and adaptability while maintaining computational efficiency. Overall, the approach is efficient in probabilistic reasoning for intelligent building automation systems.